

Concept Drift Detection and Adaptation Method for IoT Security Framework

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Abstract: With the gradual penetration of the internet of things (IoT) into all areas of life, the scale of IoT devices shows an explosive growth trend. The era of internet of everything is coming, and the important position of IoT security is becoming increasingly prominent. Due to the large number types of IoT devices, there may be different security vulnerabilities, and unknown attack forms and virus samples are appear. In other words, large number of IoT devices, large data volumes, and various attack forms pose a big challenge of malicious traffic identification. To solve these problems, this paper proposes a concept drift detection and adaptation (CDDA) method for IoT security framework. The AI model performance is evaluated by verifying the effectiveness of IoT traffic for data drift detection, so as to select the best AI model. The experimental test are given to confirm that the feasibility of the framework and the adaptive method in practice, and the effect on the performance of IoT traffic identification is also verified.

Keywords: concept drift detection and adaptive (CDDA) method; IoT security; malicious traffic identification

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I. INTRODUCTION

As the new direction of the third industrial revolution and the development of social interconnection technology in the future, the internet of things (IoT) has brought great changes to people's production and daily life after the internet development [1–5]. According to data released by authoritative statistical agencies, the number of IoT devices connected to the network worldwide reached 2.035 billion units in 2017 and will further grow to more than 7.544 billion units in 2025 [6]. However, in the booming development of the IoT devices, existing security mechanisms are unable to meet requirements of the growing security [7–9]. From the famous Mirai worm virus invading the IoT devices in 2016, which triggered a large-scale denial of service attack [10], to the recent smart scholars used by attackers to intercept users' privacy [11], the security threats of the IoT continue to emerge with the development of technology.

Traditional security issues have a special form of expression in IoT systems, which have also introduced new types of threats due to their own characteristics. Intrusion detection based on traditional labeled classification machine learning [12] is a close world problem [13]. However, the form of attack in reality is the

problem of open world without labels [14], and more and more scholars are paying attention to this research topic. On one hand, due to the many types of intelligent devices in the IoT networks, there may be different security vulnerabilities, and unknown attack forms and virus samples will appear [15]. This makes the problem of malicious traffic identification in the internet of things a typical open world problem. On the other hand, due to the large number of IoT devices, the large amount of data, and the emergence of attack forms, data anomaly detection in the field of IoT security is particularly important. In the field of IoT security, it is easier to generate conceptual drifts of data, which can cause significant harm to models and systems.

Concept drift describes how the distribution of a data stream changes unpredictably over time [16]. Lu *et al.* [17] reviewed the research results of more than 100 papers on concept drift detection, concept drift understanding, and concept drift adaptation, and points out that if the drift problem is not solved, machine learning in the concept drift environment will lead to poor learning results. Wares *et al.* [18] provided a comprehensive summary of the concept drift problem and its challenges in data stream mining. The importance of concept drift detection evaluation and the impact of dataset and time dependence on concept drift detection are discussed. Manias *et al.* [19] proposed a concept drift detection system that utilizes the federated learning updates provided in each iteration of the federated training process. Taking intelligent transportation system as an example, a framework for isolating system drift nodes based on dimension reduction and clustering technology is proposed. Data capture and drift analysis based on sliding window are combined with K-Means clustering. Jain *et al.* [20] proposed and compares conceptual drift detection based on error rate and conceptual drift detection based on data distribution. Bifet *et al.* [21] adopted a sliding window approach to distribution variations and conceptual drifts, and discusses the boundaries of false positives and false negatives. Some works [22–23] use random forest algorithms to handle different types of concept drift. Gomes *et al.* [22] proposed an effective resampling method and adaptive operator for parallel data experimentation without complex optimization of different datasets. In the literature [23], the introduction of a stream random patch algorithm is pro-

posed, and the effect of temporal diversity and average tree depth on the processing concept drift problem is further analyzed. The above-mentioned research analyzes and studies the concept drift problem from different angles, and has also achieved certain results. However, the emergence of concept drift makes it inevitable that the performance of the system will decline, and there are not many research results on how to deal with and ultimately improve the performance of the system.

Some existing works [24–26] proposed frameworks for dealing with concept drift to improve system performance. The reference [24] proposes an adaptive IoT streaming anomaly detection data analysis framework based on optimized LightGBM and concept drift adaptive, and adaptive online IoT data flow pattern changes, so that IoT data streams can be continuously learned and drift adaptive without manual intervention. The reference [25] proposes a framework for the detection and interpretation of drift samples, mapping data samples to low-dimensional spaces, and automatically learning a distance function to measure differences between samples. Use contrast learning to confirm the conceptual drift of the sample and interpret the decision boundary. In view of the concept drift problem that often occurs in IoT data analysis, the reference [26] proposed a performance-weighted probability average ensemble (PWPAE) framework for adaptive IoT anomaly detection through IoT data stream analysis. However, the above methods do not apply to unlabeled real environment.

Aiming at the security problem of malicious traffic identification in IoT, this paper proposes an concept drift detection and adaptation (CDDA) method for IoT security framework. The proposed method is evaluated by verifying the effectiveness of IoT traffic for data drift detection, so as to select the best AI model. The proposed IoT security framework starts with data preparation, using two basic datasets, NF-ToN-IoT and NF-ToN-IoT-V2. By preprocessing, the labels in the dataset are converted to two classifications, normal traffic and abnormal traffic. From the dataset NF-ToN-IoT-V2, samples are taken to generate a set of training sets and validation sets with different distributions, and from the dataset NF-ToN-IoT, samples are taken to generate a set of test sets with different distributions. During the model preparation phase, a batch of binary classification models are trained as

candidate models on each training set, and the characteristic distribution of the training set is characterized using a histogram. In the model selection phase, the histogram is used to characterize the characteristic distribution of the test set. Then, based on the similarity of the histogram, the most similar training set is found, and the training model on the training set is used as the most decision model. Through experiments, comparing multiple similarity measurement methods, it is found that the similarity measurement using Cosine and EMD is better, and the best decision model can be selected from the candidate models. The main contributions of this paper can be summarized as follows.

- Aiming at the close world problem in machine learning in the real IoT environment, we propose a data drift detection method that is suitable for open world.
- We propose a model selection framework based on the data drift detection method.
- Experimental results show that our proposed method is effective for improving the performance of IoT secure traffic awareness based on the real-world IoT traffic dataset.

II. PRELIMINARIES

2.1 Concept Drift Definition

Concept drift is an important concept in the field of data mining. The relationship between the input data and the target variable changes over time in unforeseen ways, resulting in the accuracy of the model may decrease over time [17–18]. Concept drift was first proposed by Schlimmer and Granger in 1986, stating that noise data may be converted into non-noisy information at different times, and that these changes may be caused by changes in hidden variables that cannot be directly measured [27]. Formally, concept drift is defined as follows.

Over a period of time in an observable system $[0, T]$, a subset of the data in a random sampling system at different times is recorded as $S_{(0,T)} = \{s_0, s_1, \dots, s_d, s_{d+1}, \dots, s_T\}$, where $s_i = \{X_i, y_i\}$ represents the observation value of data, X_i and y_i are the feature vector and label of the data respectively. $F_{0,t}(X, y)$ represents the joint probability distribution of random variables X and y . If there is a moment $d + 1$, satisfied $F_{0,d}(X, y) \neq F_{d+1,T}(X, y)$, concept drift is con-

sidered to have occurred. In fact, probability distributions $F_{0,T}(X, y)$ are generally difficult to obtain and are not easy to use as a basis for discriminant concept drift. Instead, use random variables for joint probability $P_d(X, y)$ judgment. If there is a moment $d + 1$, satisfied $P_d(X, y) \neq P_{d+1}(X, y)$, concept drift is considered to have occurred.

2.2 Source of Concept Drift

The joint probability of a random variable $P_d(X, y)$ can be expressed as follows.

$$P_d(X, y) = P_d(X) \times P_d(y|X), \quad (1)$$

where $P_d(X)$ represents the probability that the eigenvector of the dataset at time d is X , and $P_d(y|X)$ represents the conditional probability that the eigenvector X is identified as label y at time d . As can be seen from equation (1), $P_d(X, y)$ is composed of two parts, $P_d(X)$ and $P_d(y|X)$. Any change in either part can be regarded as a concept drift. If $P_d(X)$ changes and $P_d(y|X)$ does not change, it is called virtual drift. As long as $P_d(y|X)$ changes, it is called real drift. There are three specific cases. Since this paper mainly considers how machine learning algorithms respond to changing datasets. In order to better characterize the changes in the dataset, we divide the dataset into the above three categories. Case 1 mainly indicates that the probability distribution of the data set $P_d(X)$ has changed, but the criterion based on conditional probability $P_d(y|X)$ has not changed. Compared with Case 2 and Case 3, Case 1 is a situation that does not affect the final judgment. While Case 2 and Case 3 mainly consider the situation where the discriminant criterion changes. As can be seen from Figure 1, regardless of whether the probability distribution of the data set $P_d(X)$ changes, the judgment about the data set changes.

Case 1. $P_d(X) \neq P_{d+1}$ and $P_d(y|X) = P_{d+1}(y|X)$. As shown in Figure 1, the distribution of the data has changed, and some new sample points have emerged. But since $P_d(y|X)$ there is no change, that is, the criteria for judging the data have not changed, and the boundaries of the data classification have not changed, this is considered a false drift.

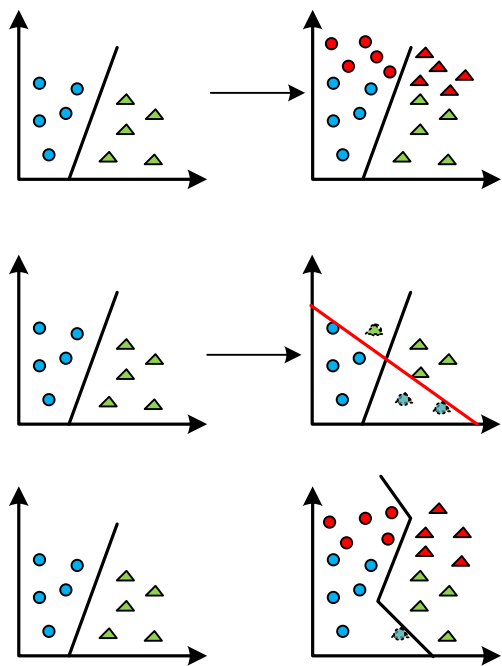


Figure 1. Three different kinds of concept drift.

Case 2. $P_d(X) = P_{d+1}$ and $P_d(y|X) \neq P_{d+1}(y|X)$. As shown in Figure 1, the distribution of the data did not change, but $P_d(y|X)$ did change, that is, the criteria for the discriminant of the data changed, and the boundaries of the data classification changed. Some of the data in the dataset is labeled differently, and the concept drift actually happens.

Case 3. $P_d(X) \neq P_{d+1}$ and $P_d(y|X) \neq P_{d+1}(y|X)$. As shown in Figure 1, the distribution of the data and $P_d(y|X)$ distribution have changed. Some new sample points have emerged, and the criteria for data discrimination have changed, and the boundaries of data classification have also changed. This is one of the most complex situations of concept drift, and in practice is the most common.

2.3 Type of Concept Drift

To better study concept drift, we divide it into four types: sudden drift, gradual drift, reoccurring drift, and incremental drift, as shown in Figure 2. A concept drift can occur not only on an exact timestamp, but also over a long period of time. Therefore, during the conversion process, an intermediate state may occur when one concept (start concept) becomes another (end concept). The differences between these concept

drift types are illustrated below.

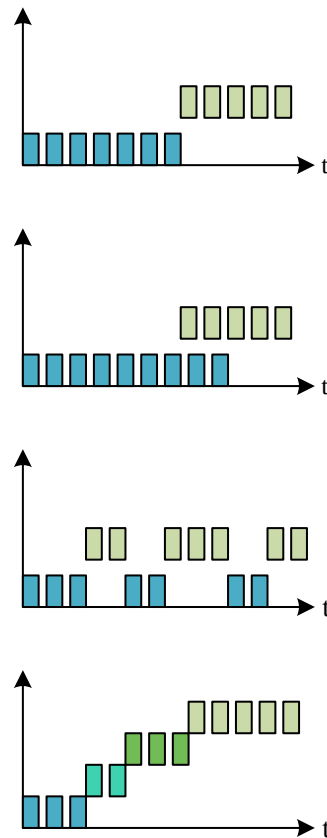


Figure 2. Four typical different types of concept drift.

2.3.1 Sudden Drift

Mathematically, the joint probability $P_d(X, y)$ of a random variable mutates at moment d . The dataset state and discriminant within the system has changed, and new models need to be replaced to accommodate concept drift. This type of drift usually occurs in emergencies and changes in legal policies. For example, the State Intellectual Property Office launched a new system at d time on a certain day of a month, requiring the application materials to adopt a new data format. Then, before time d , we use the trained model M1. After time d , there is a sudden concept drift, model M1 is not applicable, and model M2 needs to be used instead.

2.3.2 Gradual Drift

Unlike sudden drift, gradual drift refers to a gradual change in drift. The dataset state and discriminant within the system complete the change within the time

period, and model M1 is slowly replaced by model M2. Since it is a gradual drift, both model M1 and model M2 are applicable during this time period. For example, before a system comes online, a buffer is specified for a period of time, and both types of protocols can be used.

2.3.3 Reoccurring Drift

The concept drift in this type of situation occurs repeatedly, that is, the dataset state and discriminant within the system reappear after a period of time. In practice, the system adopts model M1 for a period of time, then model M2, and then model M1 over a period of time, and even periodicity and cyclical recurrence. Cyclical concept drift may be caused by seasonal and other periodic effects. For example, in the four seasons of the year, customers visit merchants and purchase records show alternating regularity.

2.3.4 Incremental Drift

Similar to gradual drift, incremental drift refers to a series of small incremental changes that occur over time. The dataset state and discriminant within the system changes over time, resulting in the need to adjust the optimization model all the time. In practice, model M1 is used at the beginning, then model M2, model M3 and so on. For example, this is the case with real-world cyber attack detection scenarios, where new forms of cyberattacks continue to emerge, and the performance of previously trained discriminative models gradually deteriorates, and then more applicable models need to be replaced.

III. THE PROPOSED CDDA METHOD

In this section, we systematically describe the proposed CDDA method, which consists of four phases: data preparation, training model candidate, model selection and predictive output. The algorithm architecture of the proposed CDDA method is shown in Figure 3.

3.1 Dataset Generation

In this paper, we first take two basic datasets, NF-ToN-IoT and NF-ToN-IoT-V2, as examples. Then, a training set and a test set are generated based on the bagging sampling method. In later chapters, the CDDA

method based on histogram and EMD distance will be further explained. Since IoT data streams are typically generated in large volumes of data continuously, learning model development using all data samples is often not feasible and unnecessary. Therefore, an efficient data sampling method should be used to select a highly representative sample of data. Prepare two IoT datasets A and B (each containing normal and abnormal traffic).

- A: Used to train the model NF-ToN-IoT-V2, the sample size 16940496;
- B: Used to test the model NF-ToN-IoT, the sample size 1379274.

Two IoT datasets A and B were sampled with playback based on bagging sampling method, and the bootstrap pseudo-samples with 812 570 samples were generated to complete the bagging sampling. Using different positive and negative sample ratios (positive sample ratio 0.1, 0.2, ..., 0.9), nine subsets A1, A2, ..., A9 (1/9 of the number of samples per subset) were obtained from dataset A through weighted random sampling.

Due to the large size of the traffic dataset in the internet of things, 5% of the original data samples were selected by the bagging sampling method to evaluate the proposed framework. The size of the subset generated can vary depending on the speed of data generation in the internet of things and the computing power of the server machine. Compared with other sampling methods, bagging sampling mainly focuses on reducing variance, so it is more effective on non-pruning decision trees, neural networks and other learners susceptible to sample disturbance.

3.2 Training Model Candidate

In the model preparation phase, a batch of binary classification models are trained as candidate models on each training set, and the characteristic distribution of the training set is characterized using histograms. Train the baseline model, select several features, train the classifier M1–M9, and generate a histogram of D1–D9 (dataset). Select 10 features of the dataset, each of which generates a histogram of equal width. For a feature, n_j denotes the frequency with which

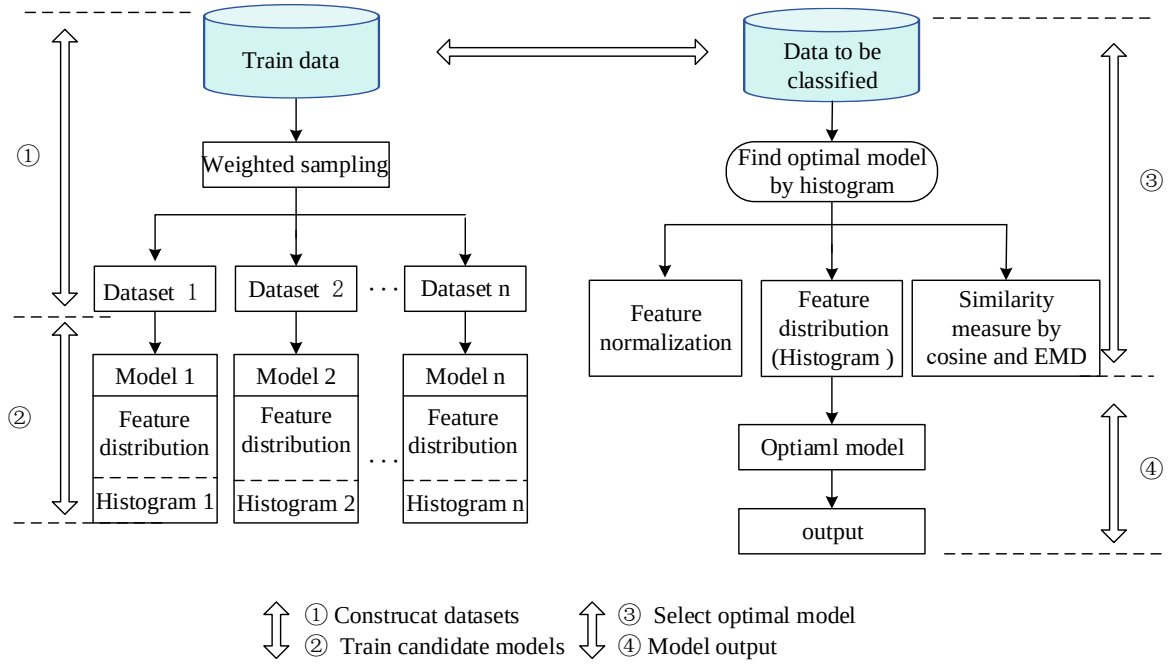


Figure 3. Algorithm framework of the proposed CDDA method.

values in the j -th bin.

$$n_j = \sum_{i=1}^N m_i, \quad (2)$$

where N represents the number of bin in the experiment, and m_i is the indicating function.

$$m_i(x) = \begin{cases} 1, & x \in \text{the } i\text{-th bin} \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

3.3 Model Selection

When processing features during the data processing phase, there will be situations where two features are not of the same magnitude. If linear operations are used, it is easy to cause imbalances. This paper uses the max-min method to normalize the features. Max-min is a linear transformation of the data, mapping the X value to between $[0, 1]$. Generally, when the characteristic data is scattered or linear and there are not many outliers, this method can be used for normalization, and the formula is as follows.

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}, \quad (4)$$

where X and X_{norm} represent the feature before and after normalization, respectively, $X_{\max}$ and $X_{\min}$ represent the maximum and minimum values of the feature, respectively. In the model selection stage, the feature distribution of the test set is characterized by a histogram, and then, according to the similarity of the histogram, the most similar training set is found, and the model trained on the training set is the most decided model. Through experiments, comparing multiple similarity measurement methods, it is found that the similarity measurement using Cosine and EMD is better, and the best decision model can be selected from the candidate models.

Earth Mover's Distance (EMD) is a 2000 International Journal of Computer Vision issue article titled The Earth Mover's Distance as a Metric for Image Retrieval proposed a method of measuring image similarity, and the concept of EMD was originally used for image retrieval. Later, because of its various advantages, it gradually used other aspects of the similarity measure. EMD itself is a hypothesis linear programming problem.

$$P = \{(P_1, \omega_{P_1}), (P_2, \omega_{P_2}), \dots, (P_M, \omega_{P_M})\}, \quad (5)$$

$$Q = \{(Q_1, \omega_{Q_1}), (Q_2, \omega_{Q_2}), \dots, (Q_M, \omega_{Q_M})\}, \quad (6)$$

where P and Q represent the feature distribution of the training sample and the test sample, respectively. P_i and Q_i represent the *bin* value of a feature i in their histograms. ω_{P_i} represents the weight of corresponding feature P_i , ω_{Q_i} represents the weight of corresponding feature Q_i . Defines the flow matrices as

$$\mathbf{F} = [f_{ij}], \quad (7)$$

where f_{ij} represents the amount of flow from P_i to Q_i . Defining d_{ij} represents the distance from P_i to Q_i to minimize the global cost function,

$$WORK(P, Q, \mathbf{F}) = \sum_{i=1}^M \sum_{j=1}^N d_{ij} f_{ij}. \quad (8)$$

Further normalized representation of the cost function is EMD distance,

$$EMD(P, Q, \mathbf{F}) = \frac{\sum_{i=1}^M \sum_{j=1}^N f_{ij}}{\sum_{i=1}^M \sum_{j=1}^N d_{ij} f_{ij}}. \quad (9)$$

Solving EMD distances is converted to solving linear programming problems of cost functions.

$$\begin{aligned} & \min \sum_{i=1}^M \sum_{j=1}^N d_{ij} f_{ij} \\ & s.t. \\ & f_{ij} \geq 0, i = 1, 2, \dots, M; j = 1, 2, \dots, N \\ & \sum_{j=1}^N f_{ij} \leq \omega_{P_i}, i = 1, 2, \dots, M \\ & \sum_{i=1}^M f_{ij} \leq \omega_{P_j}, j = 1, 2, \dots, N \\ & \sum_{i=1}^M \sum_{j=1}^N f_{ij} = \min \left\{ \sum_{i=1}^M \omega_{P_i}, \sum_{j=1}^N \omega_{P_j} \right\}. \end{aligned} \quad (10)$$

Cosine, also known as cosine similarity, is an assessment of the similarity of two vectors by calculating their angled cosine values. In this paper, we found that using cosine similarity to characterize the similar-

ity of models can also achieve good results. definition.

$$EMD(P, Q, \mathbf{F}) = \frac{\sum_{i=1}^M \sum_{j=1}^N d_{ij} f_{ij}}{\sum_{i=1}^M \sum_{j=1}^N f_{ij}}, \quad (11)$$

where n denotes the number of histograms *bin*, X denotes the training sample histogram and X_i denotes the value of i -th *bin*, Y denotes the sample histogram on the line and Y_i denotes the value of i -th *bin*, respectively.

3.4 Prediction Output

By using the histogram comparison of similarity between Cosine and EMD, the most similar training set was found, and the model trained on the training set was used to determine the model, thus overcoming the concept drift problem, improving system performance, and improving the accuracy of IoT secure traffic recognition.

IV. PERFORMANCE EVALUATION

4.1 Experimental Setup

The hardware platform is used in the experiment includes: x86_64 architecture, 20 cores, 128 GB of memory, hard disk: 256 GB; software include: scipy (1.5.2), h2o (3.34.0.3), scikit-learn (1.0.1), pandas (1.2.1), matplotlib (3.3.2), which represents the server configuration of mainstream big data analytics IoT data.

Two datasets are used to evaluate the proposed framework. The first dataset is the NF-ToN-IoT dataset, which uses publicly available pcaps to generate its NetFlow records, resulting in a NetFlow-based IoT network dataset. The total number of streams was 1 379 274, of which 1 108 995 (80.4%) were attack samples and 270 279 (19.6%) were benign samples. The second dataset is the NF-ToN-IoT-V2 dataset, with a total data stream of 16 940 496, of which 10 841 027 (63.99%) are attack samples and 6 099 469 (36.01%). Since both datasets used are unbalanced, we use four indicators, accuracy, recall and F1 score, to evaluate the overall performance of the proposed framework.

4.2 Validation Can Be Characterized

Using positive/negative sample ratios of 1:9, 2:8, 3:7,...,9:1, samples were generated into 9 training sets (ddos for abnormal traffic). Histogram and EMD was used to describe the similarity among the 9 datasets. As shown in Figure 4, the smaller the number, the lighter the color and the more similar the representation. Figure 4 visually characterizes the degree of similarity between datasets. Table 1 numerically verifies that the similarity measure is valid, and the closer the positive/negative data ratio, the more similar the results.

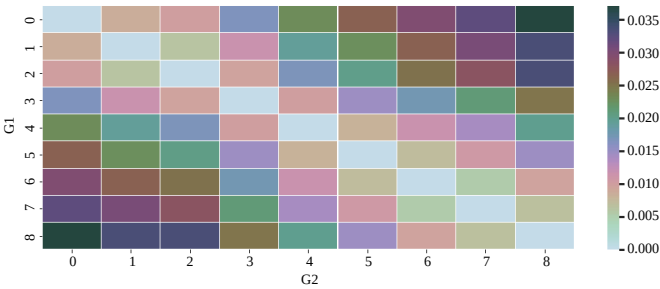


Figure 4. Histogram is used to characterize the similarity between datasets.

4.3 Comparison Performance Improvement

4.3.1 Scenes with Labels

For labeled experimental scenarios, Figure 5 depicts the average performance gain of the proposed CDDA method under labeled scenarios. Although the improvement in accuracy and precision is not obvious, i.e., 3.2% and 1.7%, while the recall and F1 score have a gain of 9.8% and 6.6%, respectively, compared to the average baseline model performance.

As shown in Table 2, the proposed CDDA method is compared with traditional Random forest (RF), Gray-Level Matrix (GLM), Gradient Boosting Machine (GBM), DEEPLearn, Convolutional Neural Network (CNN) and Bayes models, respectively. Although individual performance is not as good as RF and DEEPLearn, it is essentially caused by the overfitting of these two models and is not applicable to open world problems in real-world scenarios. On the other hand, although individual performance is not optimal, the average performance of the proposed CDDA method is better than the six models mentioned above.

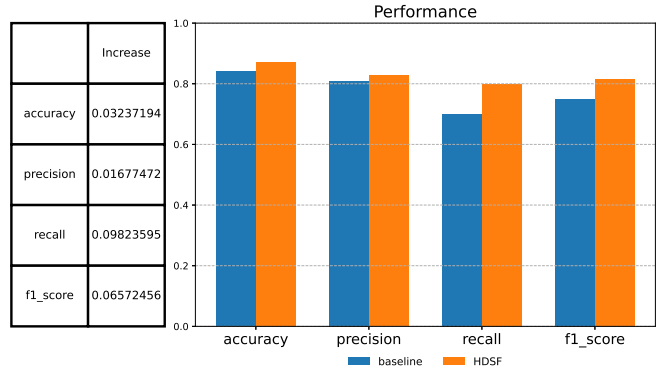


Figure 5. Overall performance improvement.

Table 3 compares the overall performance of the proposed CDDA method in a labeled scenario with traditional RF, GLM, GBM, DEEPLearn, CNN, and Bayes models.

4.3.2 Real Environment Without Labels

For labeled experimental scenarios, Figure 6 depicts the average performance gain of the proposed CDDA method in unlabeled scenarios. Compared with Figure 5, the overall improvement is small, especially for the performance indicator of accuracy. But unlabeled scenarios are more realistic application environments, and the proposed CDDA method gives an exploratory effort. Although the improvement in accuracy and precision is not obvious, there are 7.4% and 4.6% gains in recall and F1 score, respectively.

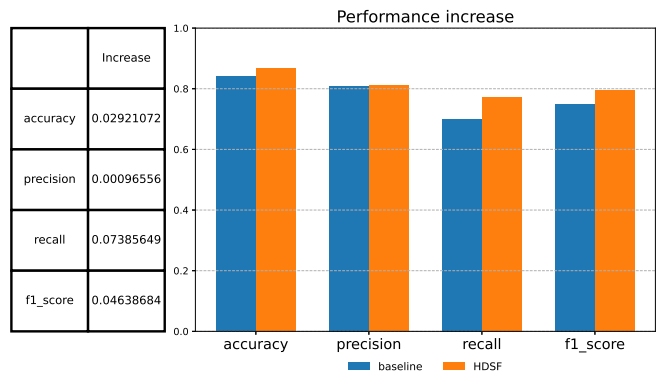


Figure 6. Overall performance improvement.

As shown in Table 4, the proposed CDDA method is compared with traditional RF, GLM, GBM, Deeplearn, CNN and Bayes models, and the performance improvements are significant, except for small accuracy improvements. Similarly, the improvement

Table 1. Uses histograms to characterize similarities between datasets.

G1/G2	0	1	2	3	4	5	6	7	8
0	0.000000	0.008596	0.010063	0.016540	0.022981	0.026756	0.029817	0.032284	0.037067
1	0.008594	0.000000	0.006249	0.011592	0.011592	0.022691	0.026750	0.030440	0.033431
2	0.010065	0.006250	0.000000	0.009690	0.016933	0.020124	0.025228	0.028201	0.033381
3	0.016541	0.011587	0.009695	0.000000	0.009921	0.014731	0.017503	0.021120	0.025076
4	0.022976	0.019321	0.016934	0.009921	0.000000	0.008165	0.011707	0.014094	0.019846
5	0.026744	0.022687	0.020127	0.014718	0.008161	0.000000	0.007003	0.010671	0.014764
6	0.029814	0.026742	0.025223	0.017496	0.011708	0.007022	0.000000	0.005297	0.009623
7	0.032282	0.030436	0.028197	0.021114	0.014093	0.010670	0.005300	0.000000	0.006607
8	0.037065	0.033427	0.033369	0.025072	0.019841	0.014757	0.009622	0.006609	0.000000

Table 2. Elevation of each model.

	Accuracy	Precision	Recall	F1 score
RF	−0.00012769	−0.000347093	−0.00141618	−0.00242901
GLM	0.12639792	−0.00952783	0.25691671	0.15286328
GBM	0.02667125	−0.00116768	0.05314280	0.02668254
Deeplearn	−0.0212379	−0.01275931	−0.00141750	−0.00715475
CNN	0.02502224	0.13489374	0.27278561	0.22206910
Bayes	0.01839170	−0.00731969	0.00940429	0.00231622
Average	0.03237194	0.01677472	0.09823595	0.06572456

Table 3. Overall performance improvement (general map): Labeled scenarios.

	Baseline				CDDA (proposed)				Increase			
	Accuracy	Precision	Recall	F1 score	Accuracy	Precision	Recall	F1 score	Accuracy	Precision	Recall	F1 score
RF	0.95752698	0.98242932	0.92341394	0.95187126	0.95739929	0.9789581	0.92199776	0.94944225	−0.0001277	−0.0034709	−0.0014162	−0.002429
GLM	0.77975988	0.89143825	0.59964024	0.71397053	0.9064578	0.88191042	0.85655695	0.86683381	0.12639792	−0.0095278	0.25691671	0.15286328
GBM	0.95067177	0.97691151	0.91250709	0.94338573	0.97734302	0.97574383	0.96564988	0.97006827	0.02667125	−0.0011677	0.0531428	0.02668254
Deeplearn	0.94280228	0.98313609	0.89344675	0.93603929	0.94067849	0.97037678	0.89202926	0.92888453	−0.0021238	−0.0127593	−0.0014175	−0.0071548
CNN	0.49242348	0.10745819	0.00131005	0.00251619	0.51744572	0.24235193	0.27409566	0.22458529	0.02502224	0.13489374	0.27278561	0.2220691
Bayes	0.94389635	0.94361632	0.85416156	0.89551208	0.93217805	0.93629663	0.86356535	0.8978283	0.0183917	−0.0073197	0.00940429	0.00231622
Average	0.83949513	0.81416495	0.69741327	0.74054918	0.871864706	0.83093966	0.79564923	0.80627374	0.03237194	0.01677472	0.09823595	0.06572456

Table 4. Elevation of each model.

	Accuracy	Precision	Recall	F1 score
RF	0.00113408	−0.00321378	0.00189657	−0.00051347
GLM	0.07158803	−0.02927362	0.15969995	0.07537674
GBM	0.03230318	−0.00197413	0.06375612	0.03277711
Deeplearn	0.00070333	0.00302191	0.00155628	0.00223637
CNN	0.04232198	0.04145876	0.16595822	0.14079421
Bayes	0.02721373	−0.00422575	0.05027182	0.02765012
Average	0.02921072	0.00096556	0.07385649	0.04638684

Table 5. Overall performance improvements (overview): Unlabeled scenarios.

	Baseline				CDDA (proposed)				Increase			
	Accuracy	Precision	Recall	F1 score	Accuracy	Precision	Recall	F1 score	Accuracy	Precision	Recall	F1 score
RF	0.95278543	0.98372349	0.9144947	0.94995287	0.95391951	0.98050971	0.91639127	0.9472394	0.00113408	−0.0032138	0.00189657	−0.0005135
GLM	0.844056	0.91078779	0.73230034	0.80963842	0.91564403	0.88151416	0.89200029	0.88501516	0.07158803	−0.0292736	0.15969995	0.07537674
GBM	0.93847576	0.97823459	0.88887478	0.9312572	0.97077894	0.97626046	0.95263089	0.96403432	0.03230318	−0.0019741	0.06375612	0.03277711
DEEPLearn	0.93547946	0.98344919	0.88053067	0.92903295	0.9361828	0.9864711	0.88208695	0.93126931	0.00070333	0.00302191	0.00155628	0.00223637
CNN	0.48491836	0.09010857	0.00077841	0.001522	0.52724034	0.13156732	0.16673663	0.1423162	0.04232198	0.04145876	0.16595822	0.14079421
BAYES	0.87497562	0.94163116	0.77915949	0.85167756	0.90218935	0.93740541	0.82943131	0.87932768	0.02721373	−0.0042257	0.05027182	0.02765012
Average	0.83844844	0.8146558	0.6993564	0.74514683	0.8676916	0.81562136	0.77321289	0.79153368	0.02921072	0.00096556	0.07385649	0.04638684

is not significant because the two models, RF and Deeplearn, are trained to fit, but in fact they cannot be used in labeled real-world scenarios. On the other hand, the average performance of the proposed CDDA

method is significantly better than that of GLM, GBM, CNN and Bayes. The average performance of the proposed CDDA method is better than the six models mentioned above. Table 5 compares the overall perfor-

mance of the proposed CDDA method with traditional RF, GLM, GBM, Deeplearn, CNN, and Bayes models in a unlabeled scenario.

V. CONCLUSION

In this paper, we aim at the open world problem in the identification of malicious traffic in the internet of things, and propose an IoT security framework and algorithm for concept drift detection and adaptation. By comparing the similarity histogram features of Cosine and EMD, the most similar training set was found. The model trained on the training set is the most decision model, which overcomes the concept drift problem, improves the system performance and improves the accuracy of IoT traffic identification. In future research work, exploring the histogram features of characterizing datasets and improving the Cosine and EMD distances are all directions worthy of in-depth research.

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